



Beyond Searching a Village: Learning to Recommend Diverse and Successful Collaborative Teams

<https://fani-lab.github.io/OpeNTF/tutorial/recsys26/>



fani-lab.github.io



recsys.acm.org/recsys26/



TIME AND LOCATION

ABSTRACT

TARGET AUDIENCE

OUTLINE

PIONEERING TECHNIQUES

NEURAL TEAM RECOMMENDATION

TIME AND LOCATION

Half-day, Afternoon, 1:30 PM - 5:30 PM (GMT/UTC -6), Monday, September 28, 2026

Elk/Deer Lake (4th floor), Marriott City Center, Downtown, Minneapolis, Minnesota, USA

Prev Tutorials at UMAP24, SIGIR-AP24, and WSDM25, and CIKM25 < [Full Outline] [Slides] [Recording]

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[Team Projects List](#)[1. Project](#)

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[Definition and](#)[Team Formation](#)[2. Background and](#)[Related Work](#)

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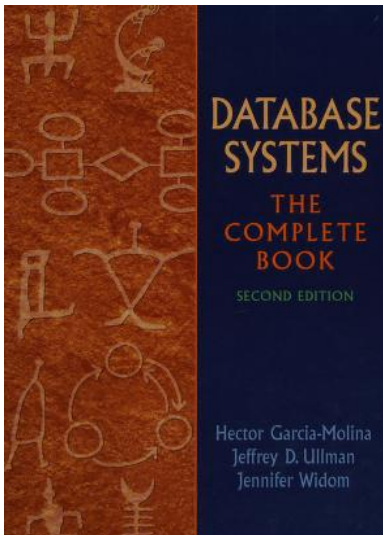
Team Projects List

[Print](#)[Settings](#)

Add dates and restrictions...

Team Members		Project
	Taimoor Shahzad	
1	Mason Li	Extending the Financial Industry Business Ontology to Represent Time-Varying Exchange-Traded Fund Alloca
	Muhammad Shahmeer Ather Kakwan	
	Avlyn Kaur Khurana	
2	Harish Pugazhenthii Sasikala	Modeling a Multi-Provider Vehicle Retail Ecosystem for Interoperable Incentive Reasoning
	Krishnaswamy Gopalakrishnan	
	Hanif Muktadi	
3	Dev Malik	Towards Interoperable Flood Information Across South Asia
	Ata Sanoubar	
	Meet Shah	
4	Sakshi Pinkeshkumar Patel	Modelling Football Match Events and Player Connections: Bayer Leverkusen 2023/24
	Ayush Bharatbhai Patel	
	Jungwon Youn	
5	Sunghun Cho	PluggedIn: Representing Electric Vehicle Charging Accessibility
	Elias Dockal	
	Karan Jony Brahmhatt	
6	Moksh Patel	Screenplay-to-Screen: An Interoperable Knowledge Representation for Film Casting and Production Networks
	Asish Achenkunju George	
7	Khaled Abuela	TRIAD: Trusted RDF Interoperable Agent Data
	Aung Htut	
8	Bowen Zhang	OrcaLink: Modeling Potential Social Ties in Southern Resident Killer Whales
	Deepa Nelgi	
9		A Data Modeling Approach to BC Transit Services

Outline: W02-03



CH04 (2nd Ed.)

- I) Entity
- II) Attribute
- III) Relationship
- IIII) Extended ER (EER)

Hands-On



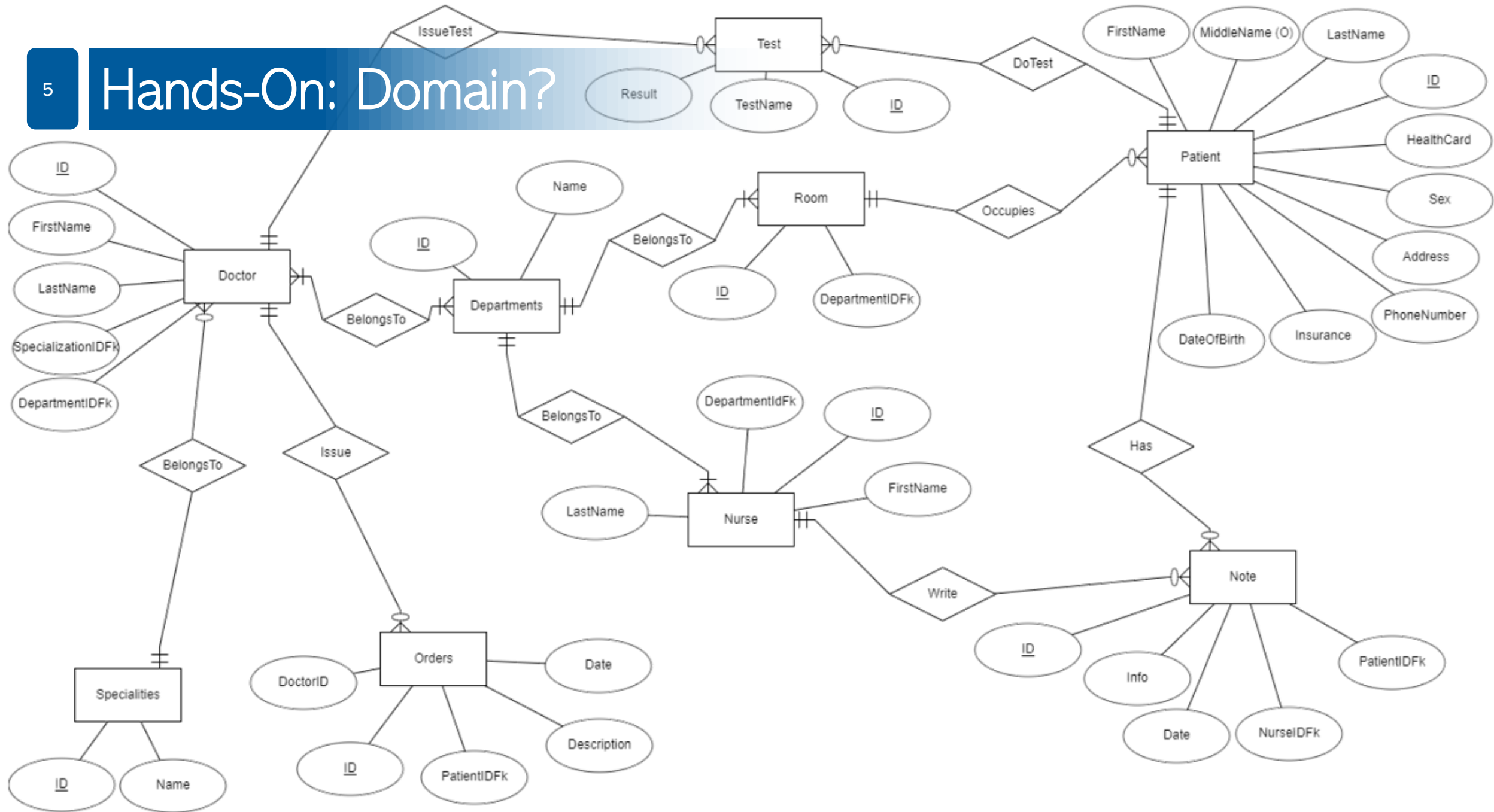
F1 Boxenstopp I-IV, 2007

by Andreas Gursky

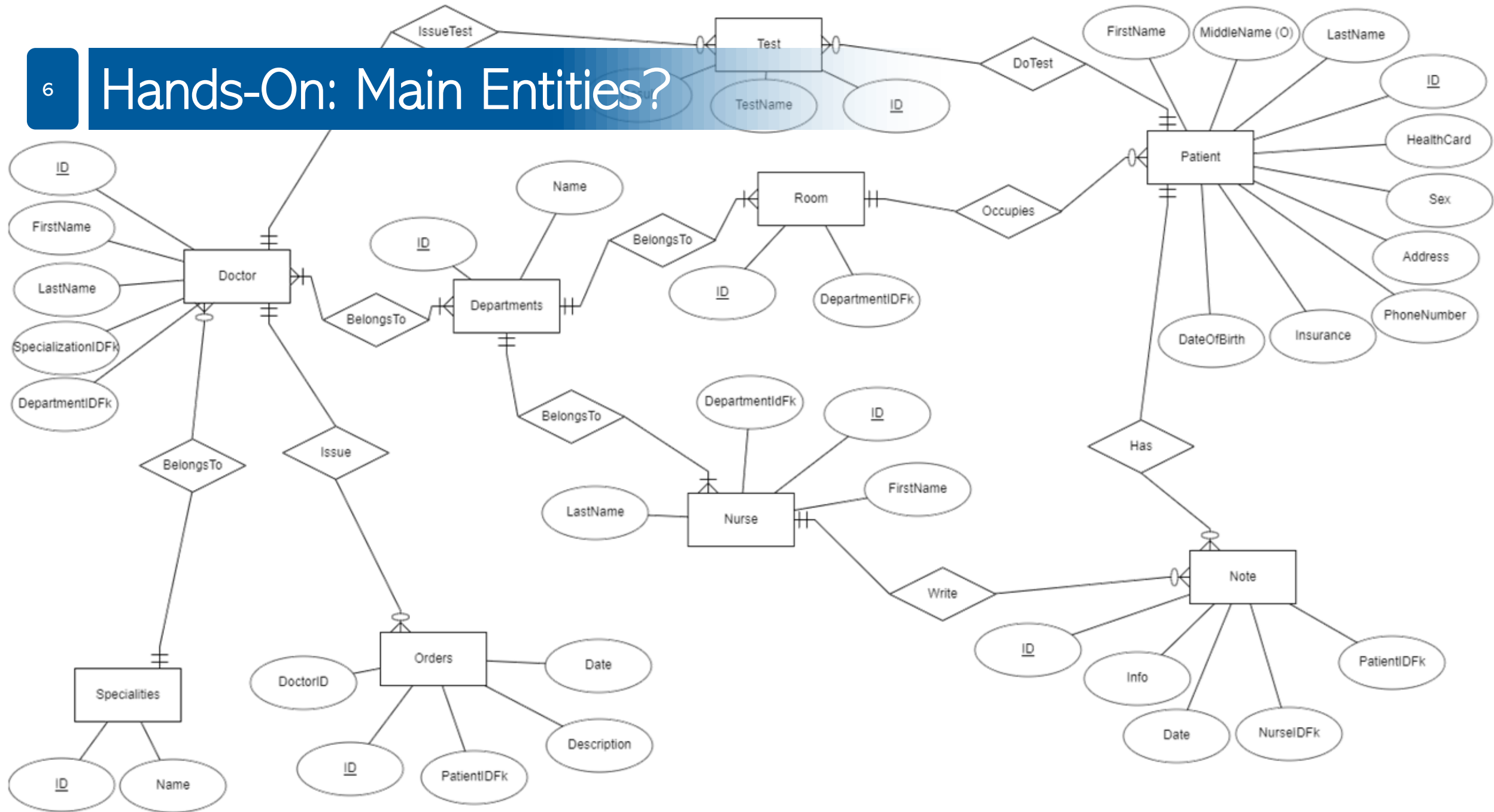
https://en.wikipedia.org/wiki/Rhein_II



Hands-On: Domain?

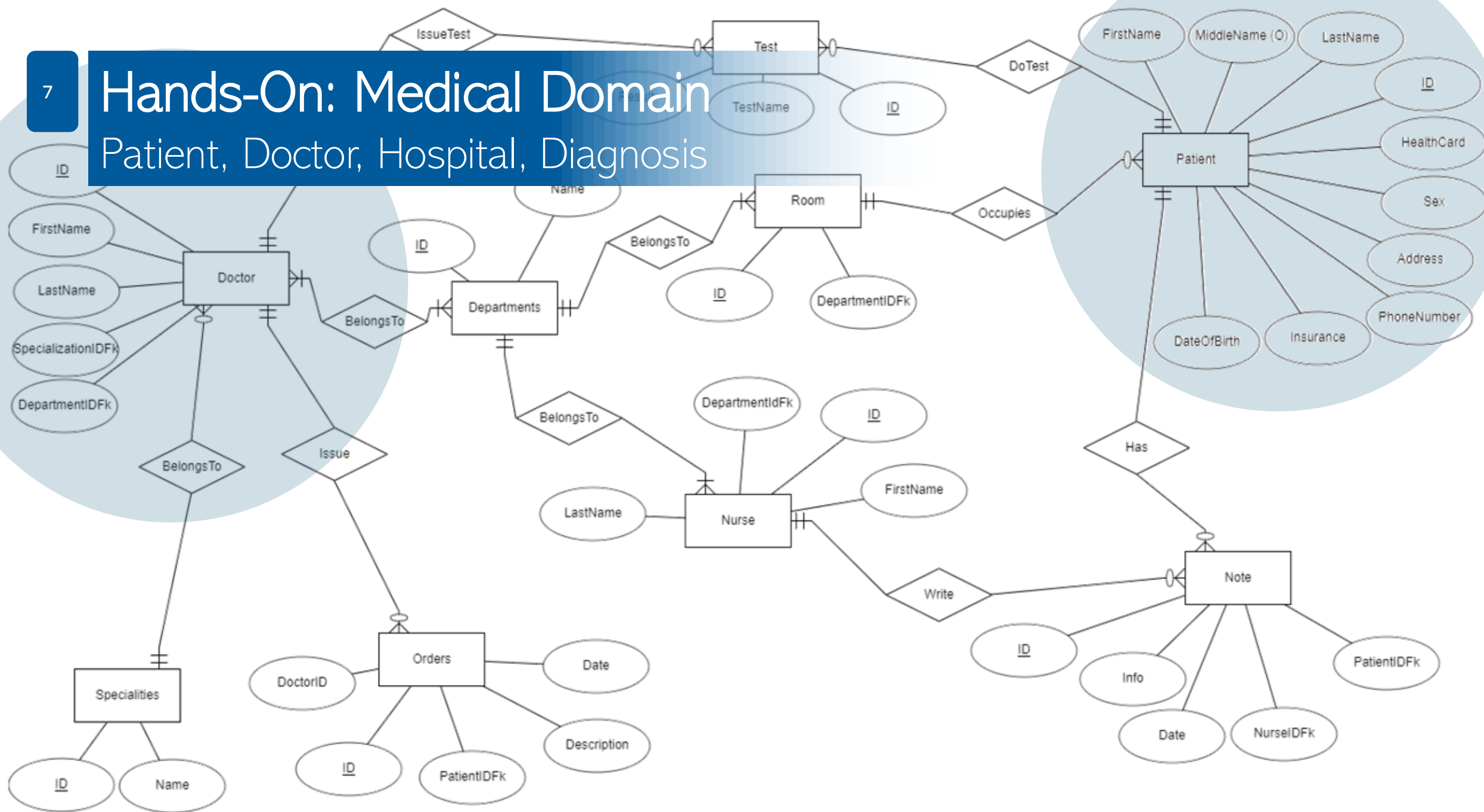


6 Hands-On: Main Entities?



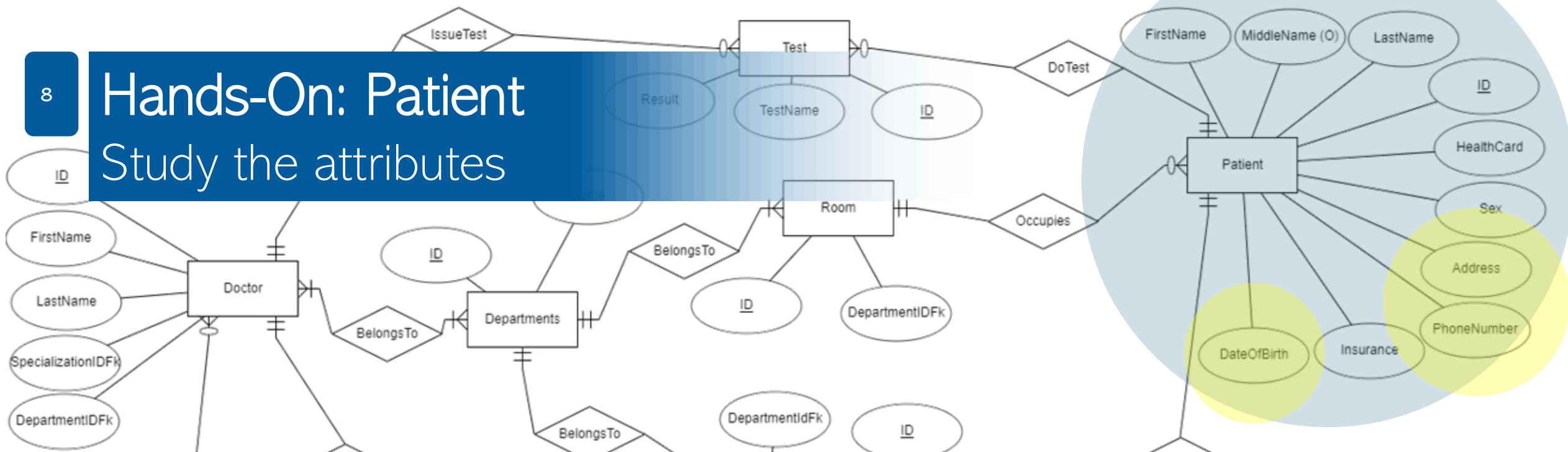
Hands-On: Medical Domain

Patient, Doctor, Hospital, Diagnosis



Hands-On: Patient

Study the attributes

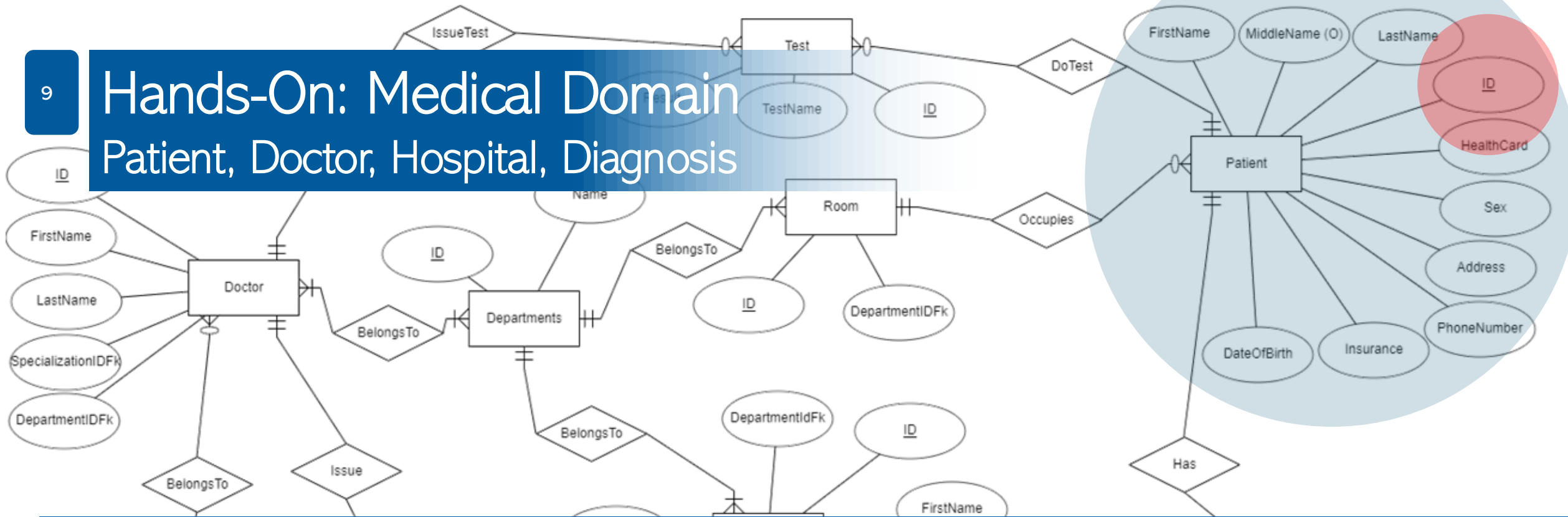


Address, PhoneNumber, and DateOfBirth as atomic attributes vs. composite.
The reason might be the fact that the subparts:
AreaCode in PhoneNumber is not essential for this system.
If needed, the subparts can be extracted at the Application Level (Programming Level)



Hands-On: Medical Domain

Patient, Doctor, Hospital, Diagnosis



Is this a key in real-world, or an autogenerated key to keep entities unique?

In hospitals, a patient is identified by a unique number, but it is not called ID. ID commonly known as Passport, ID Cards, ...

Let's rename it to Number. Also, not PatientNumber since the entity name explains it.

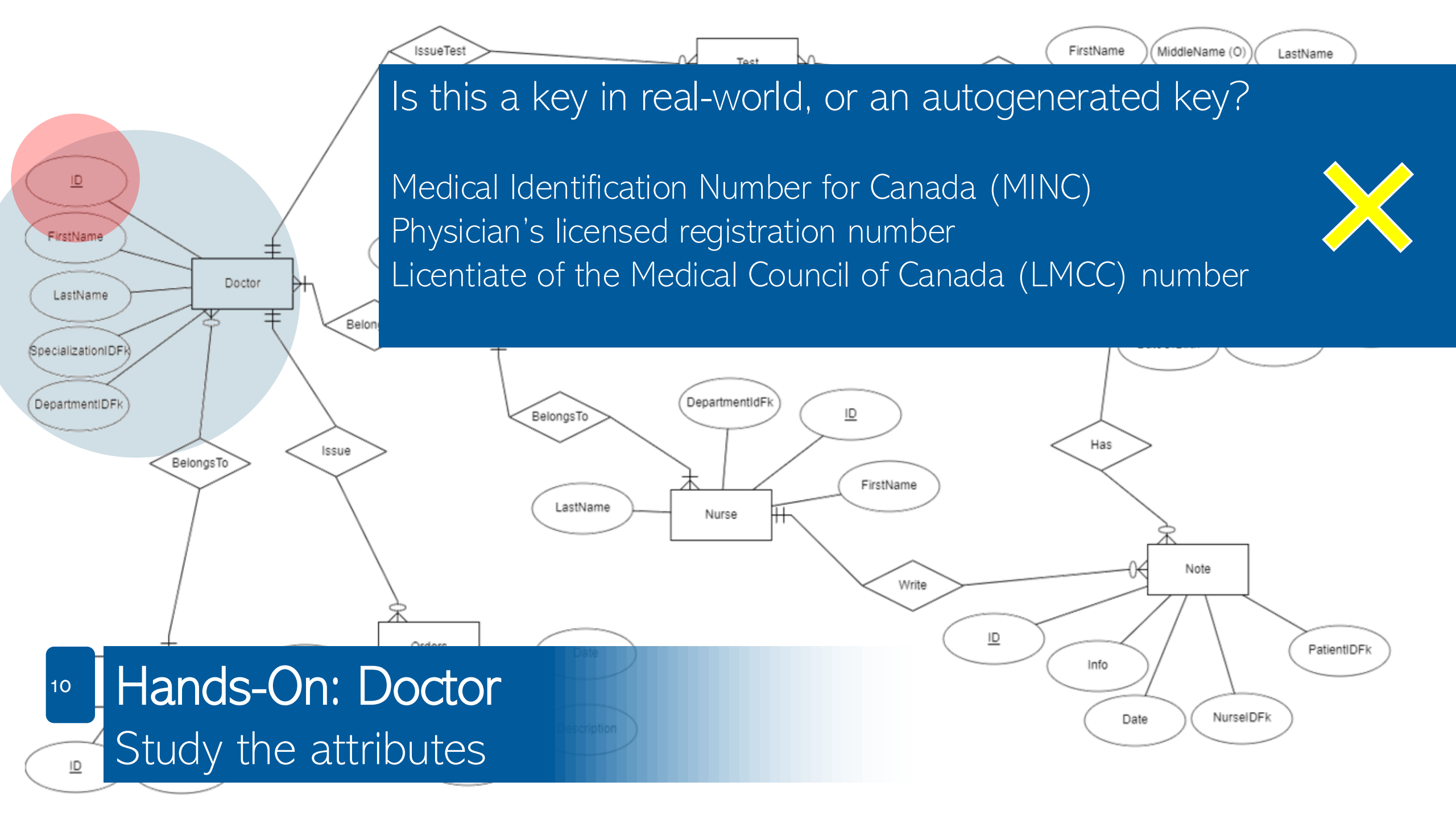


Is this a key in real-world, or an autogenerated key?

Medical Identification Number for Canada (MINC)

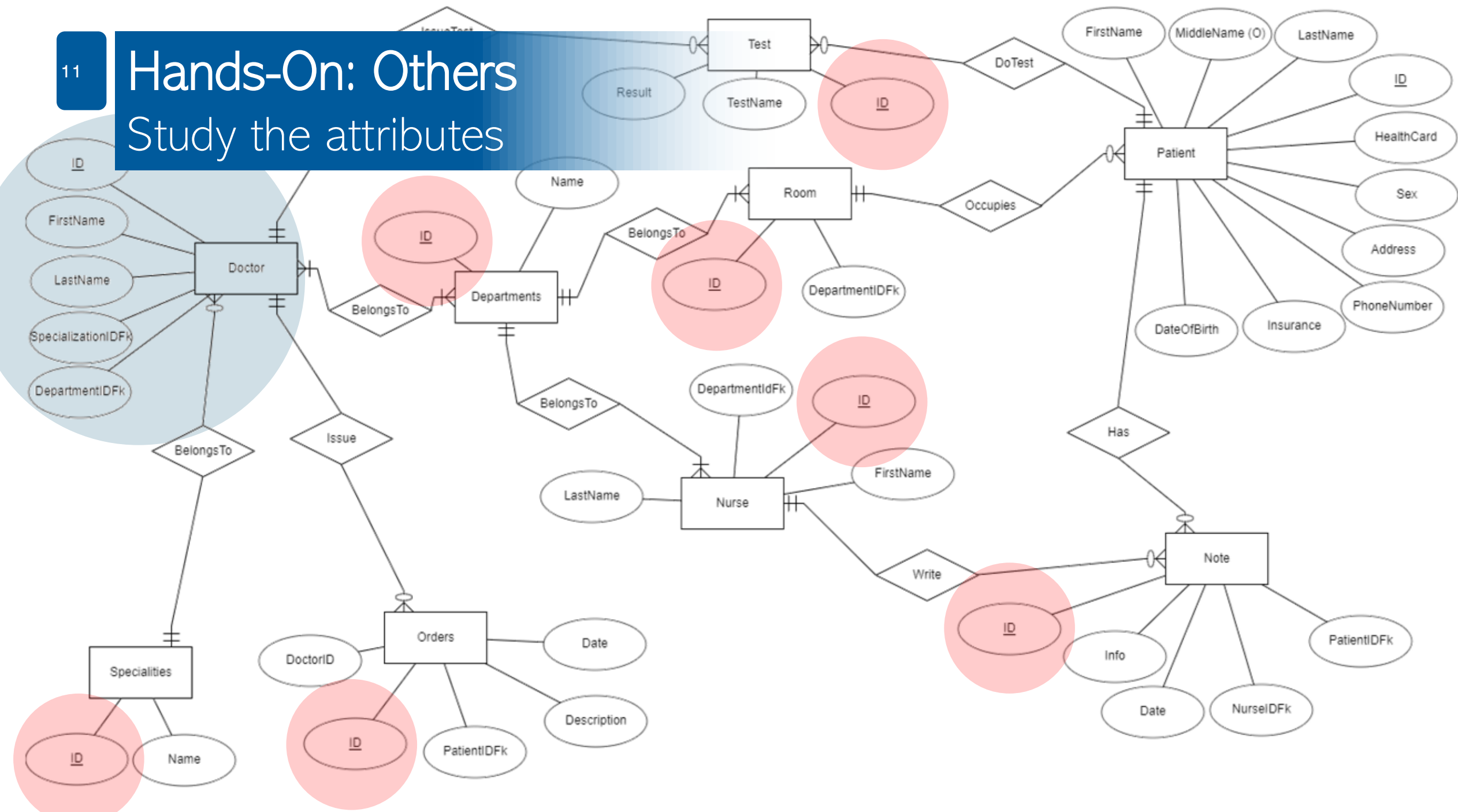
Physician's licensed registration number

Licentiate of the Medical Council of Canada (LMCC) number



Hands-On: Others

Study the attributes



Is this a Foreign Key (FK)?

DepartmentIDFk, SepcializationIDFk

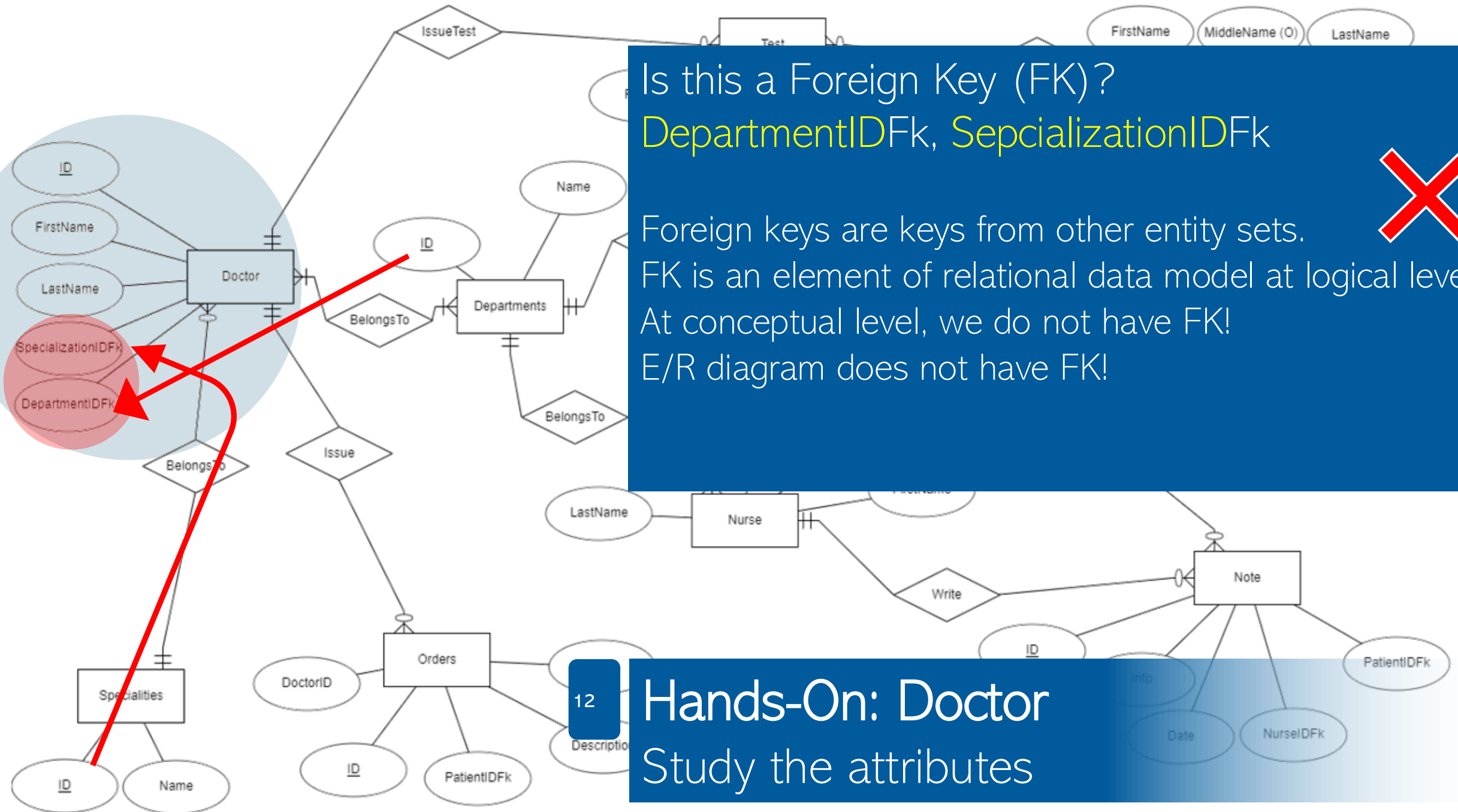


Foreign keys are keys from other entity sets.

FK is an element of relational data model at logical level.

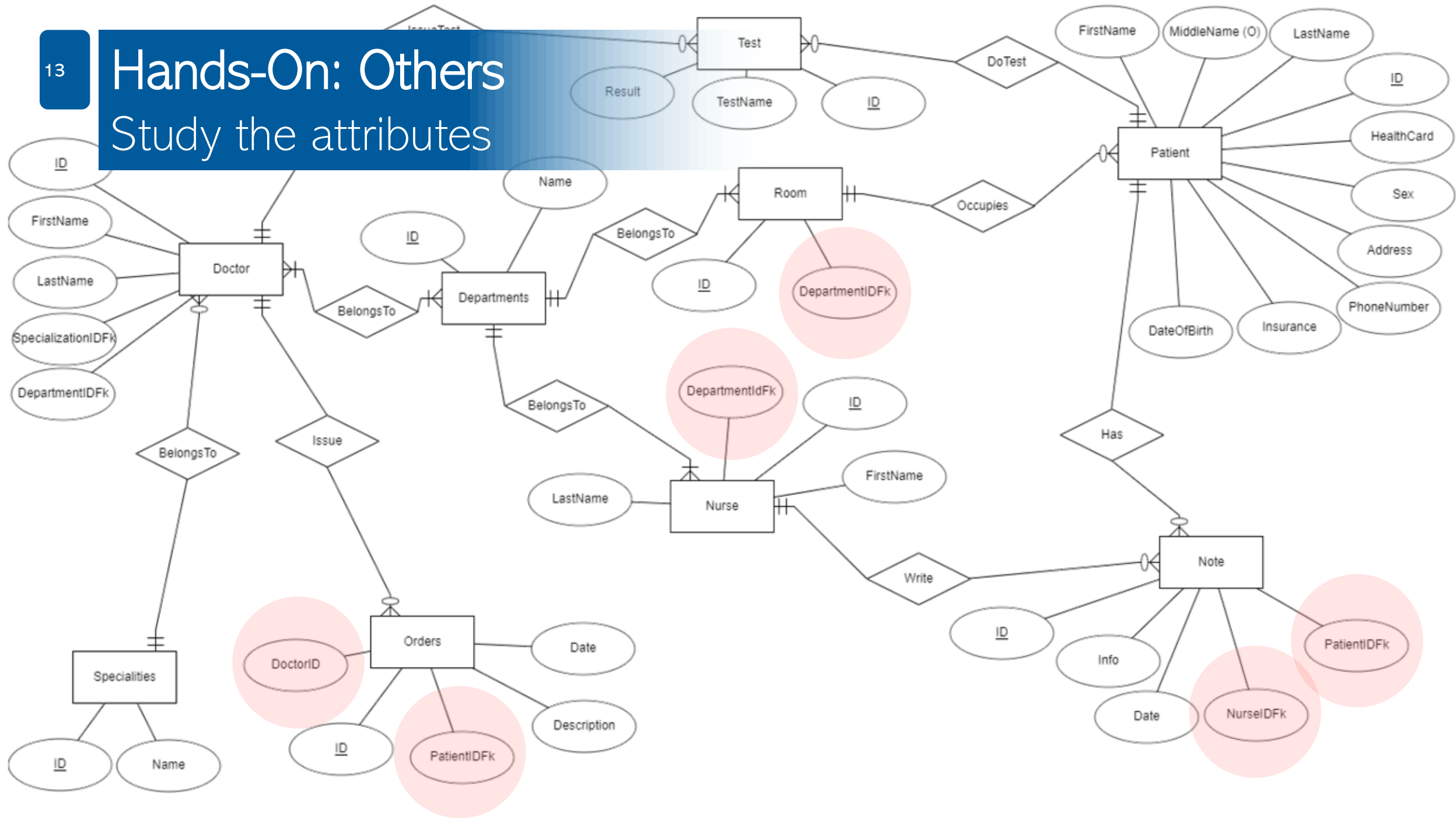
At conceptual level, we do not have FK!

E/R diagram does not have FK!



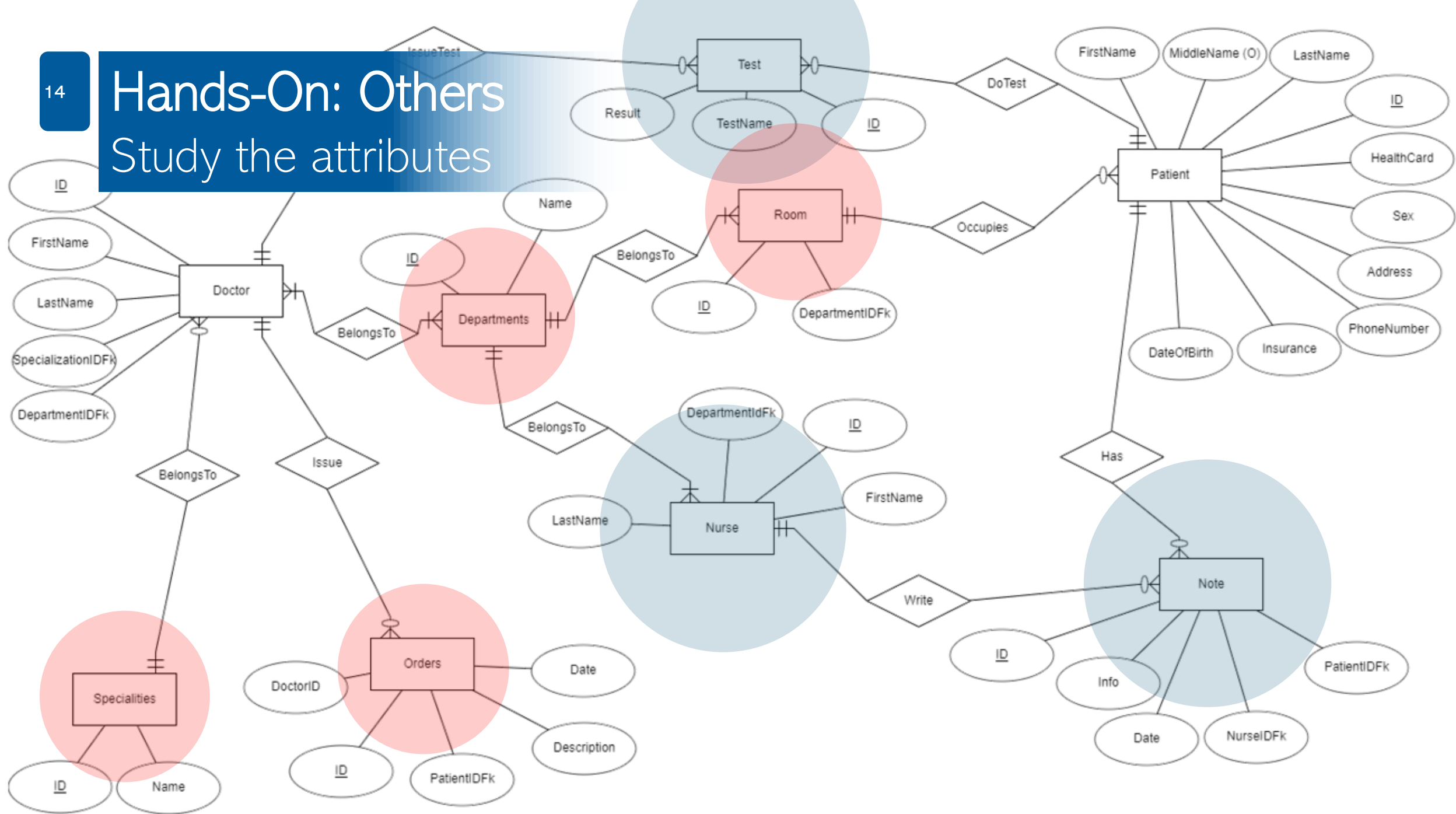
Hands-On: Others

Study the attributes



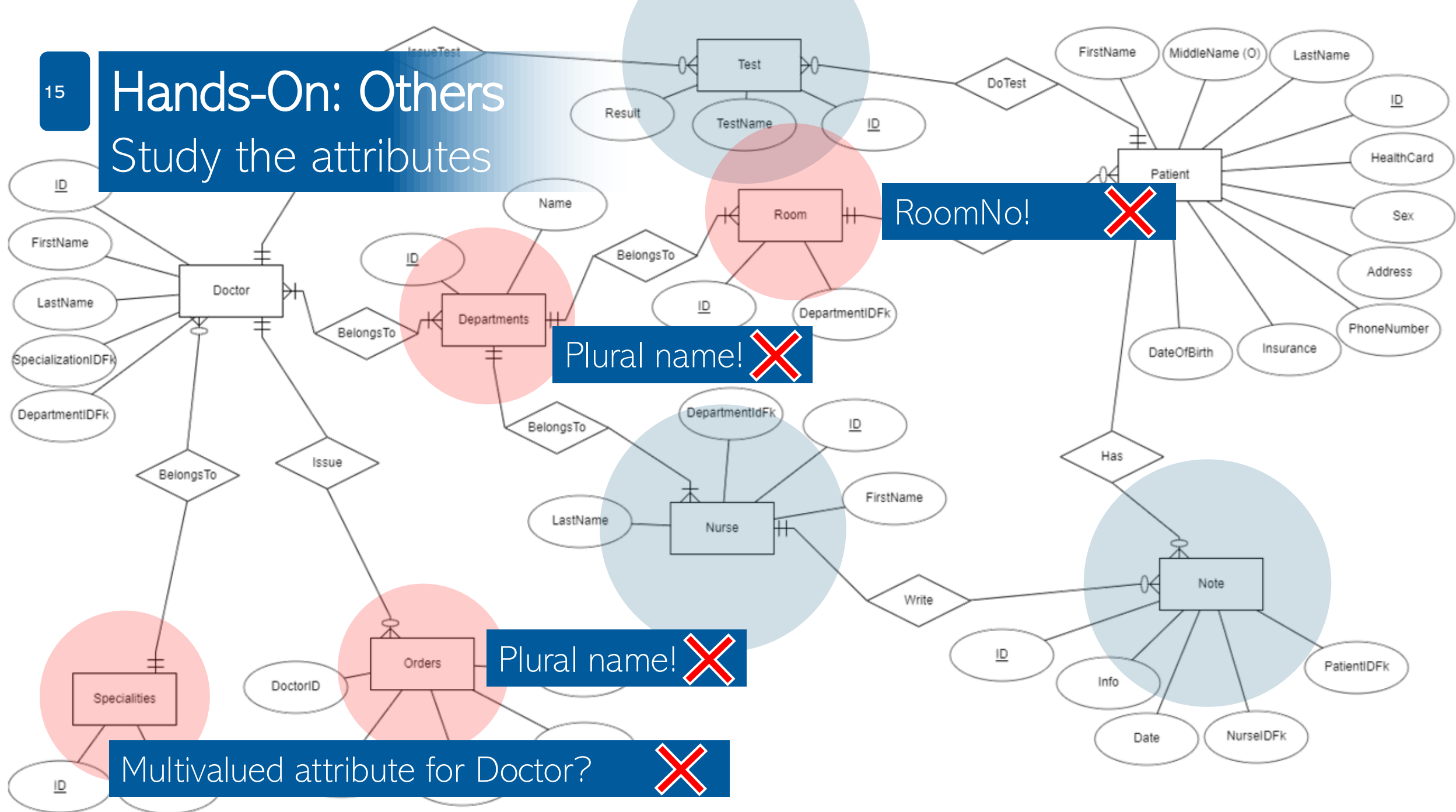
Hands-On: Others

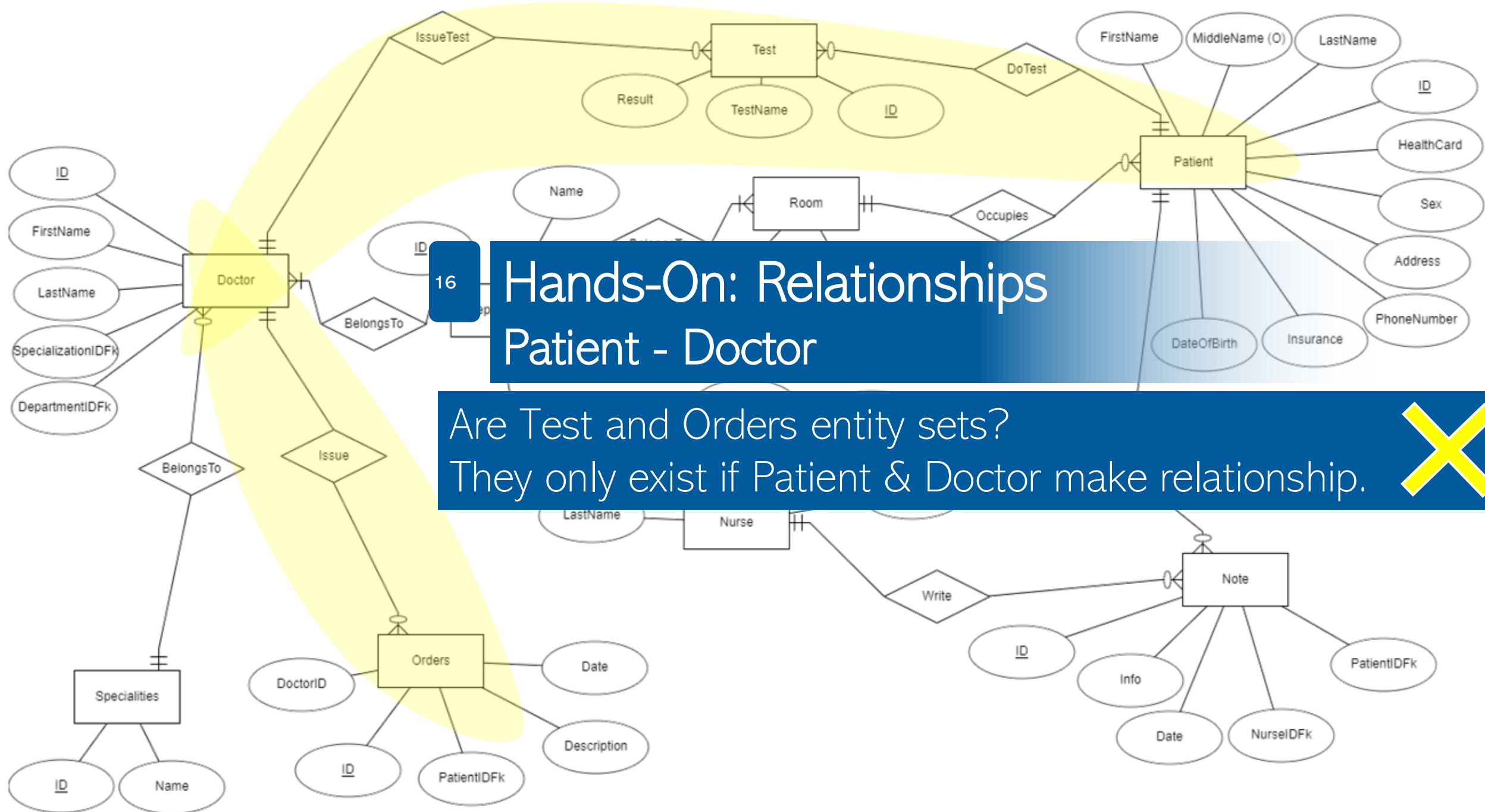
Study the attributes



Hands-On: Others

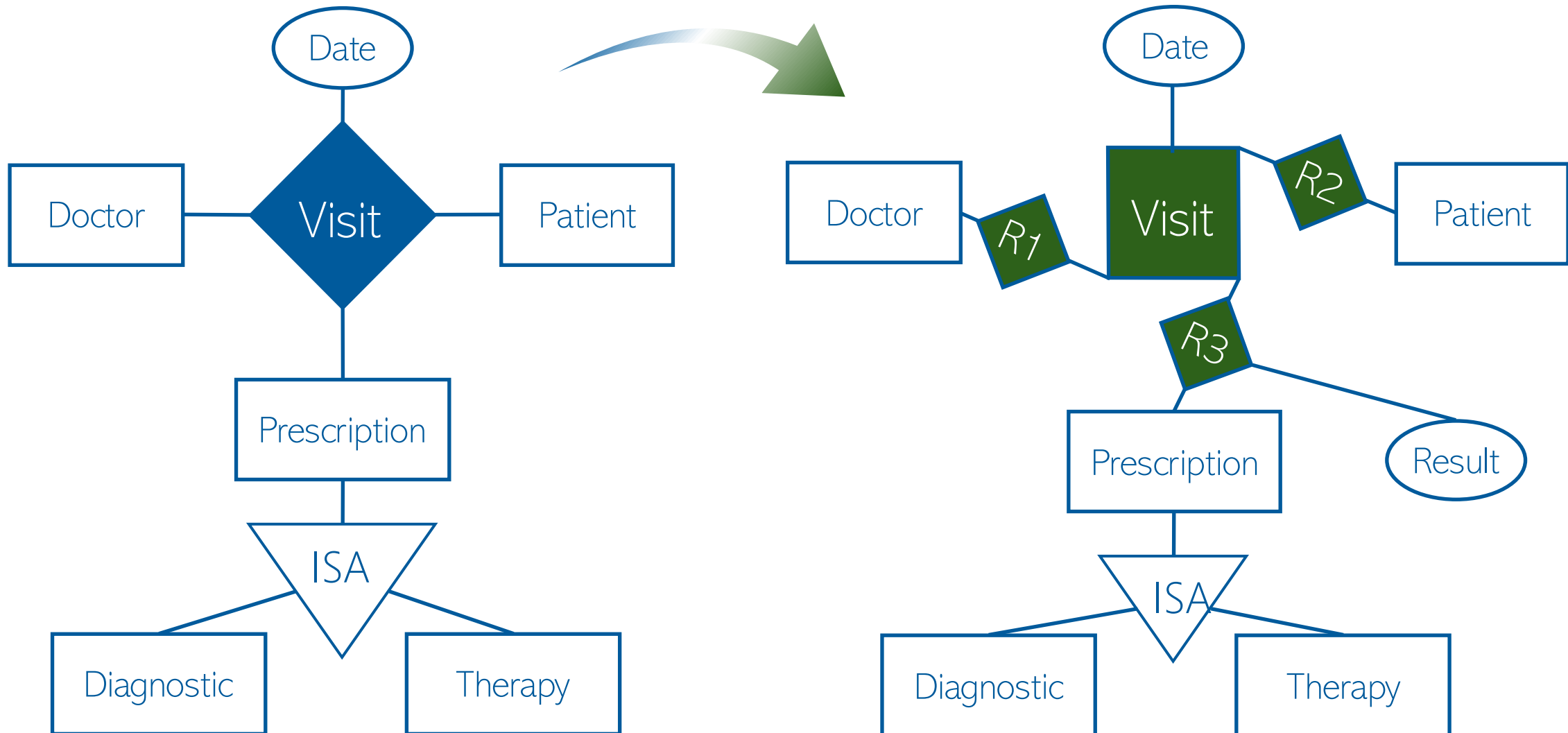
Study the attributes

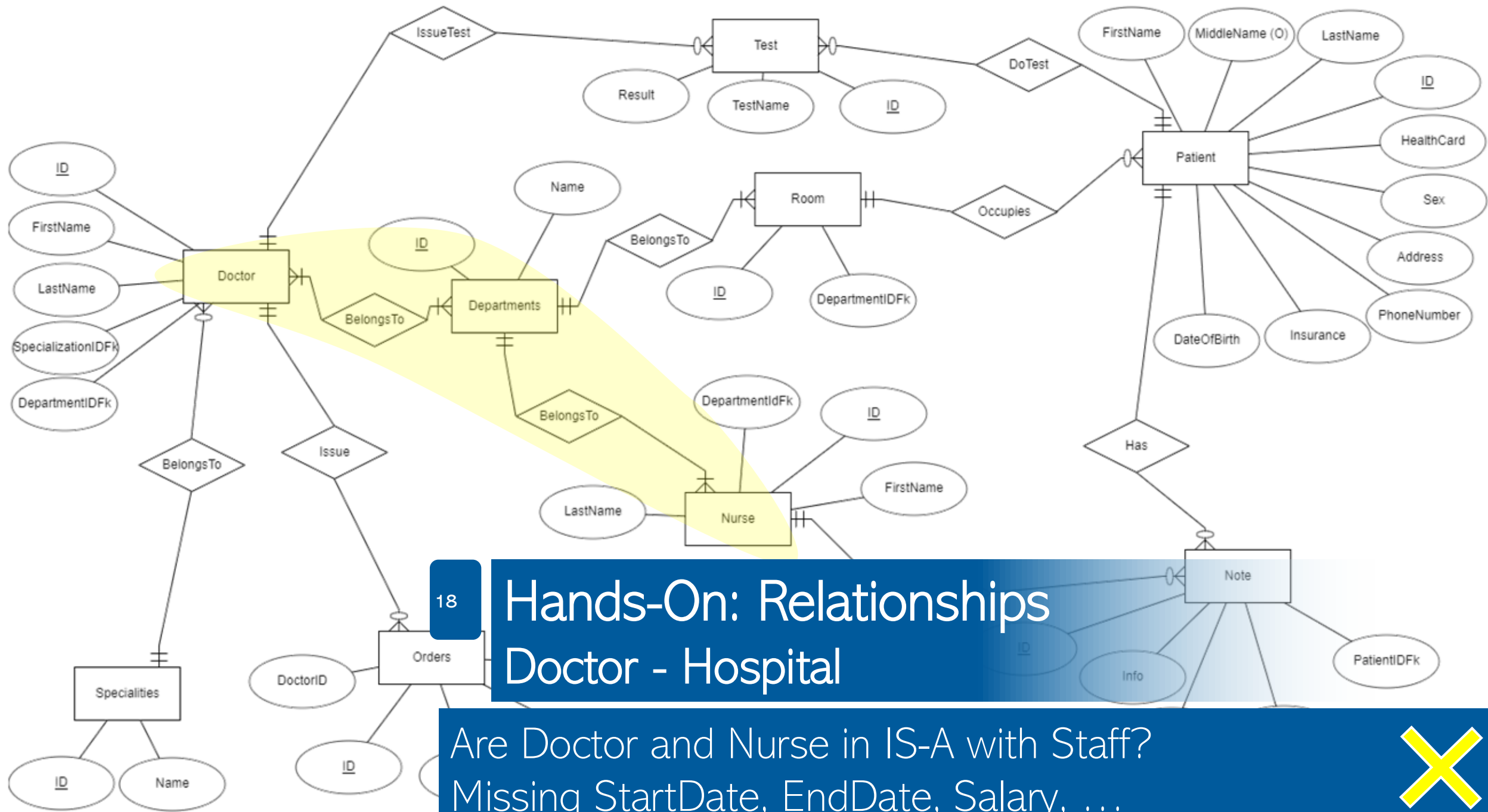




Hands-On: Relationships

Patient - Doctor



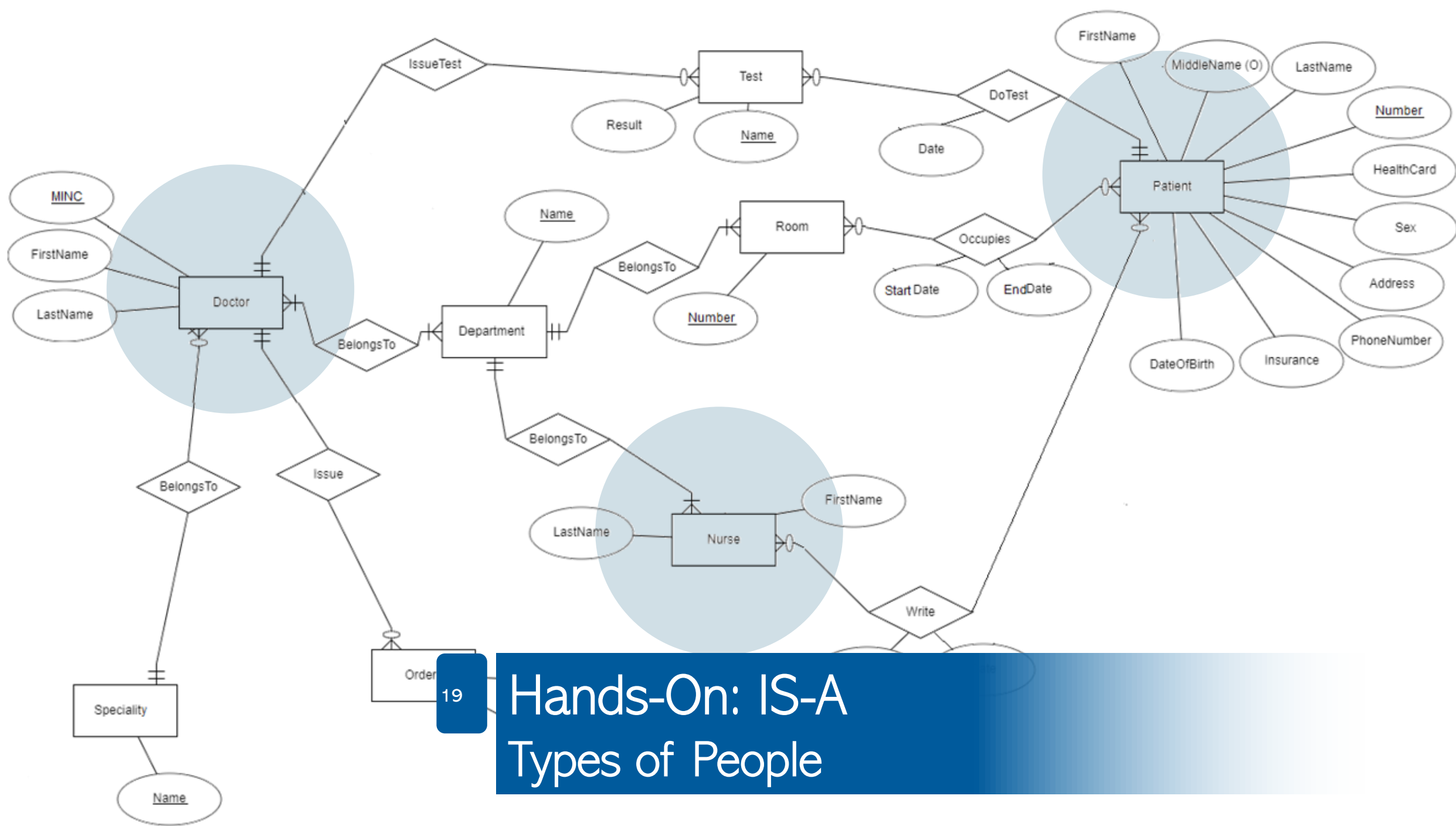


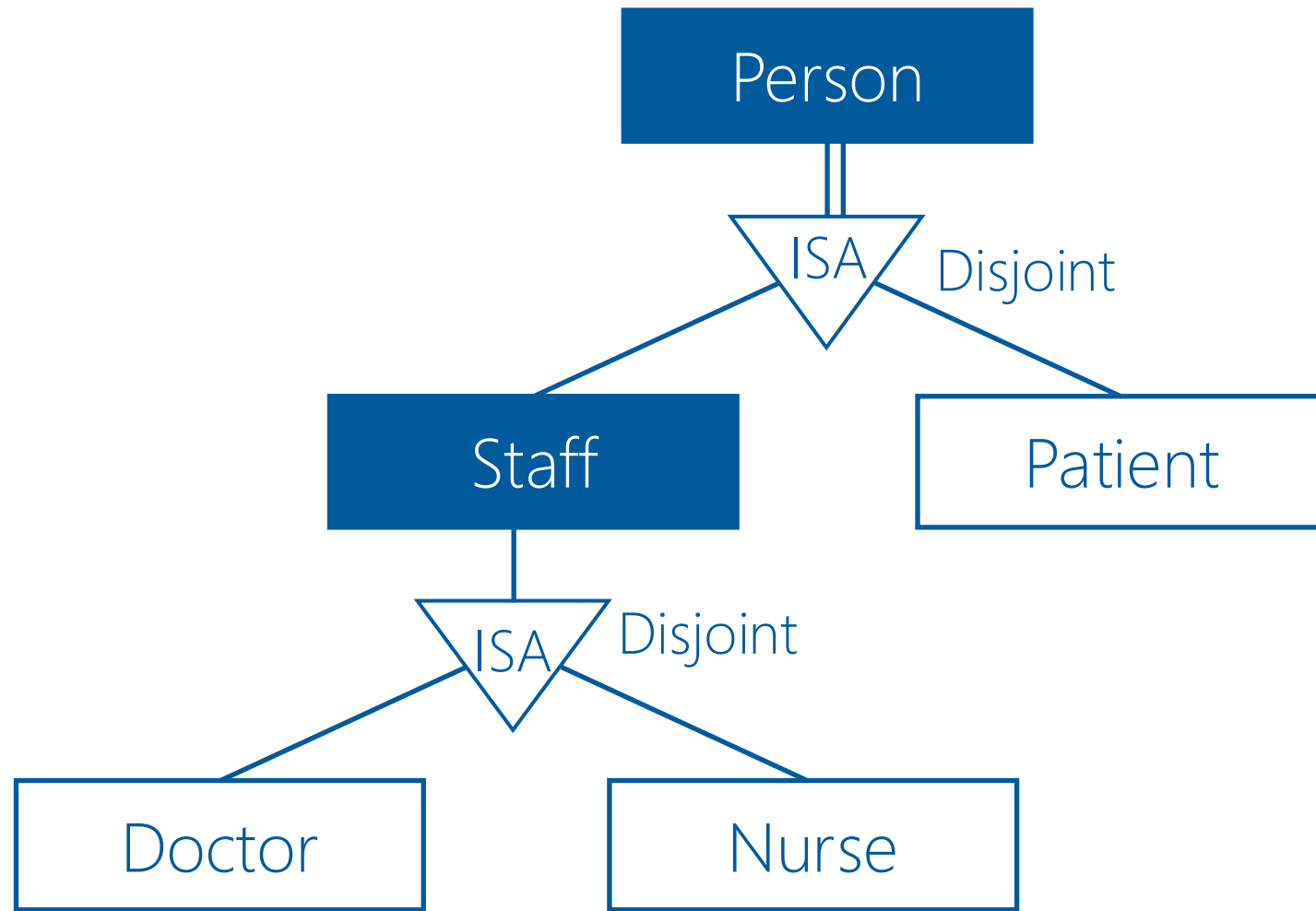
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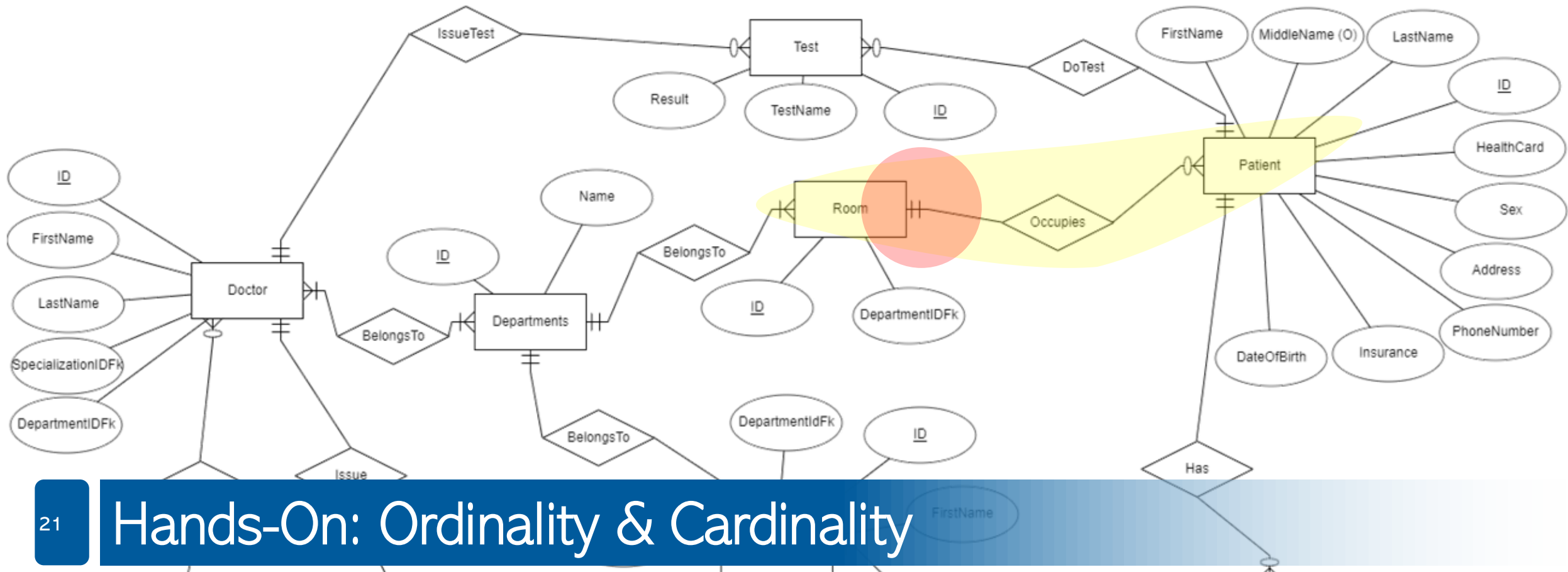
Hands-On: Relationships Doctor - Hospital

Are Doctor and Nurse in IS-A with Staff?
Missing StartDate, EndDate, Salary, ...









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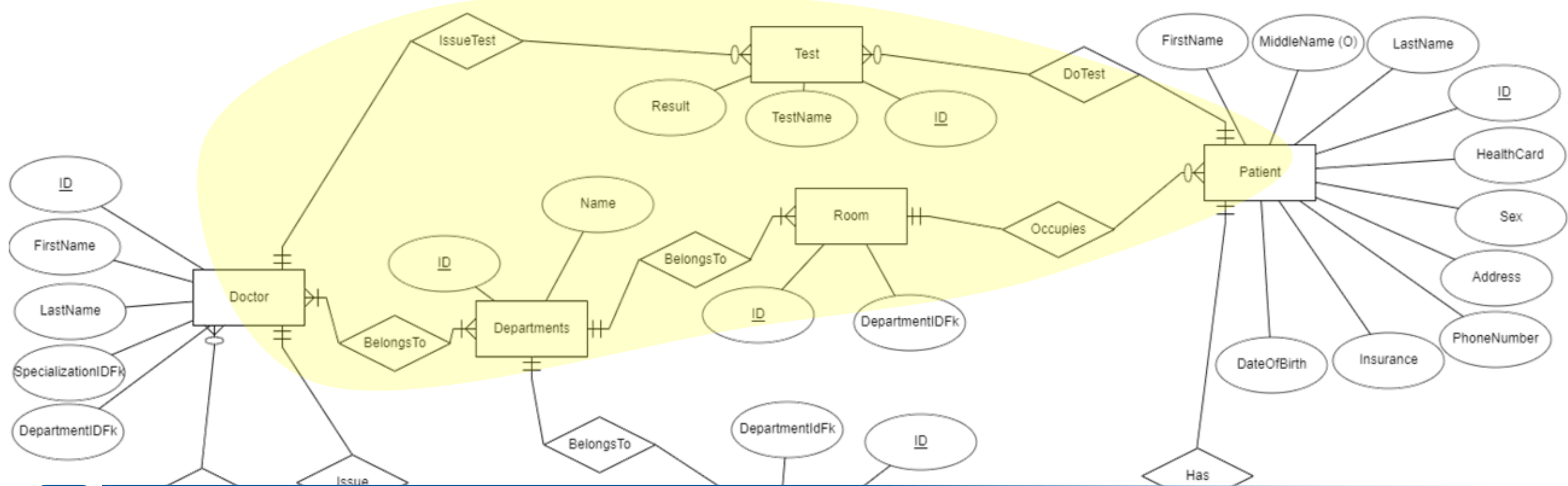
Hands-On: Ordinality & Cardinality

A Patient entity must be in a room!

Also, what happens if Patient leaves the hospital.

Also, what happens if Patient are different rooms during stays.

Also, how to track each stay in a room, missing StartDate & EndDate

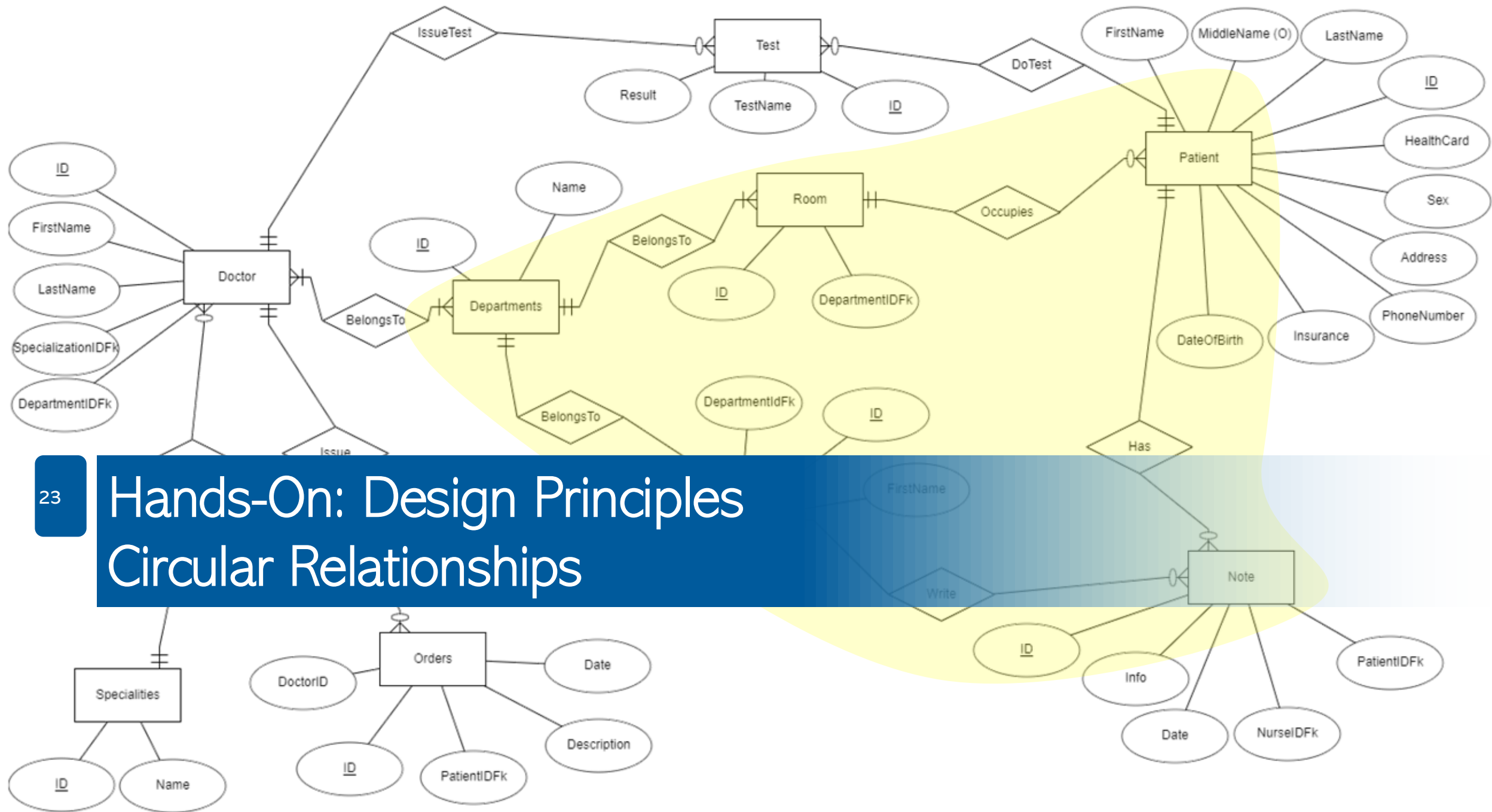


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Hands-On: Design Principles

Circular Relationships







F1 Boxenstopp I-IV, 2007
by Andreas Gursky

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A Data Odyssey: Relational DBMS vs. GGG



Data Rep.
in
RDBMS

Real World Entity

Conceptual Level | Entity-Relationship Model Level

| Logical Level | Relational Model

| Physical Level | SQL

| Computable Entity



Data Rep.
in
GGG

Real World Entity

Conceptual Level | Web Ontology Language (OWL)

| Logical Level | RDF & Description Language

| Physical Level | SPARQL

| Computable Entity

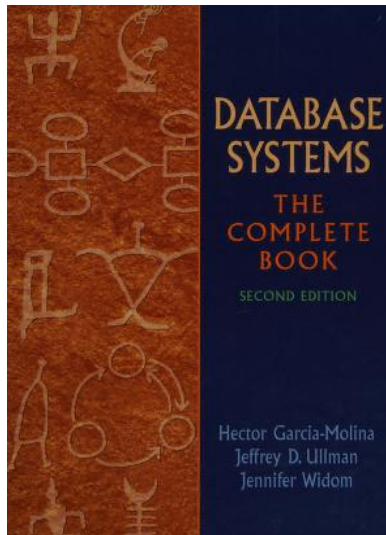
Don't be squared ..., Pulp Fiction, Quintine Tarantino, 1994



Logical Level | Relational Data Model



Outline: W04

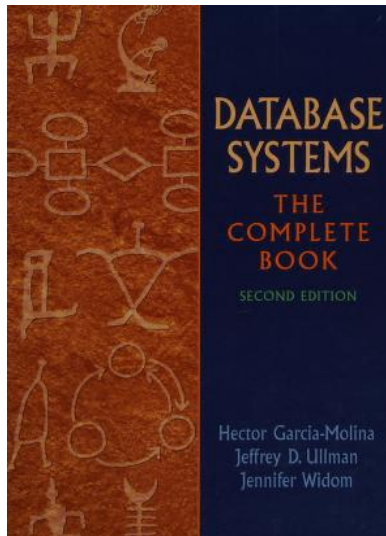


CH02 (2nd Ed.)

- I) Relational Modeling
- II) Entity to Relation
- III) Relationship to Relation
- III) IS-A to Relation

Hands-On

Outline: W04A



CH02 (2nd Ed.)

I) Relational Modeling

II) Entity to Relation

III) Relationship to Relation

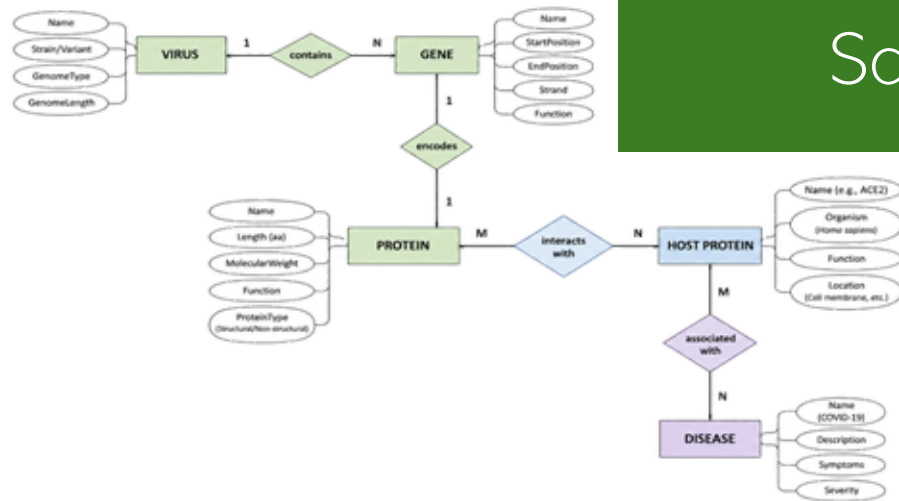
IIII) IS-A to Relation

Hands-On

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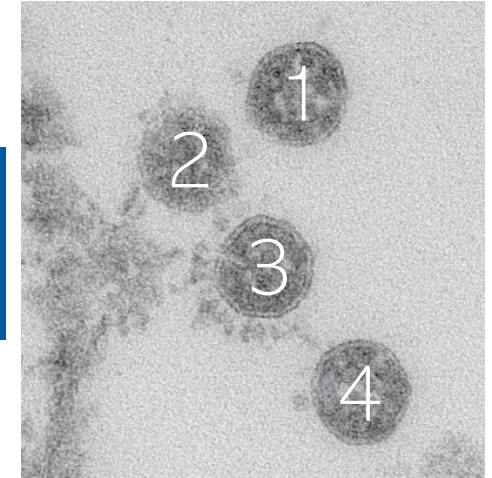
Abstractions: Logical Representation

How **instances** of entities, **values** of their attributes, their relationships should be represented based on a schema.



Schema

Real-World
Instances of Entity



Abstractions: Logical Representation

Common logical models include:

~1960: Object Oriented

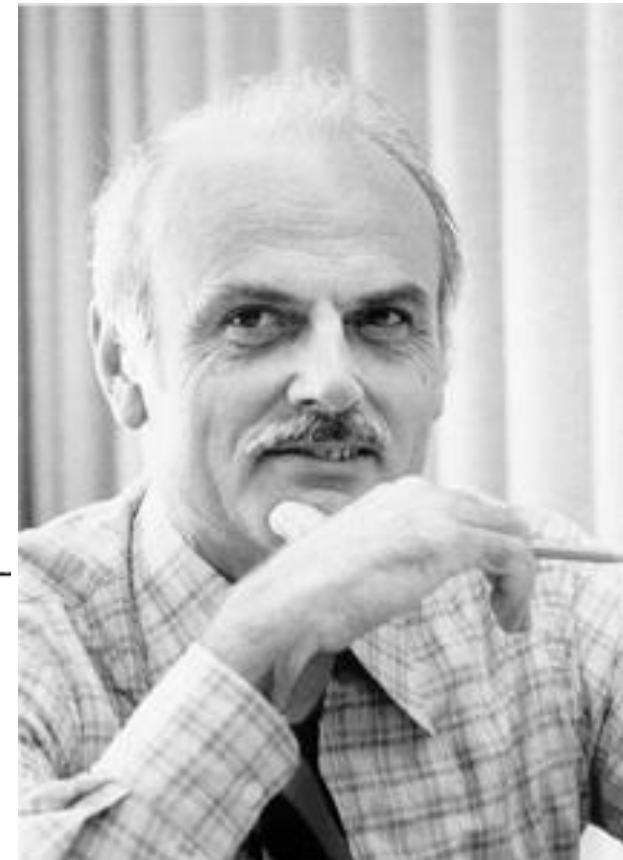
~1969: Relational: Mathematical Relation

~1996: XML

Edgar Frank "Ted" Codd, IBM, 1969, 1970
Information Retrieval

**A Relational Model of Data for
Large Shared Data Banks**

E. F. CODD
IBM Research Laboratory, San Jose, California

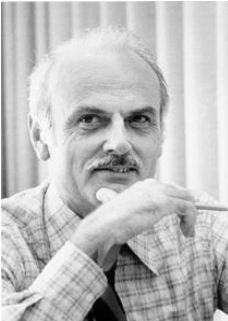


Logical Model: Relational

In math, if $A, B, C, \dots$, are sets, the Cartesian product is the set of all tuples $(a, b, c, \dots)$ such that:

$$A \times B \times C \times \dots = \{(a, b, c, \dots) \mid a \in A, b \in B, c \in C, \dots\}$$

A **Relation** is a subset of Cartesian product.



Logical Model: Relational

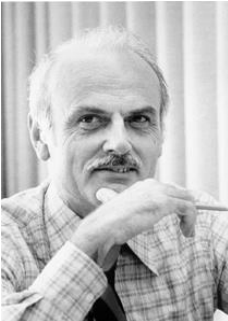
An Entity Set with Attributes coming from domains $A, B, C, \dots$, is like,

$$A \times B \times C \times \dots = \{(a, b, c, \dots) \mid a \in A, b \in B, c \in C, \dots\}$$

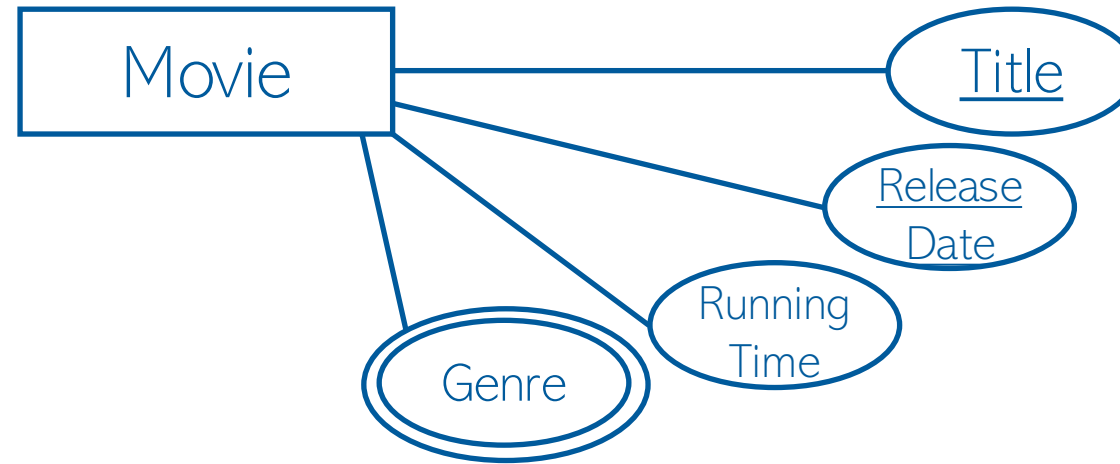
that can include all possible entities of the Entity Set.

→ Data instance is then represented as a **Tuple**: $(a, b, c, \dots)$

→ Entity Set is then a **Relation**: $\{(a, b, c, \dots), (a', b', c', \dots), \dots\}$



Relational Model: Entity Set is Relation



Title = {a, aa, aaa, ... all sequences of chars of variable length}

ReleaseDate = {1890, 1891, ..., 2026}

RunningTime = {1, 2, 3, ..., 240}

Genre = {Horror, Western, Drama, Comedy, ...}

Title × ReleaseDate × RunningTime × Genre = {(a, b, c, d) | a ∈ Title, b ∈ ReleaseDate, c ∈ RunningTime, d ∈ Genre}

(Rosemary's Baby, 1968, 136, {Horror, Drama}), (2001: A Space Odyssey, 1968, 139, {SciFi, Drama}), ...

Relational Model: Algebra

Algebra:

- A set of operations

- A set of elements

- A set of axioms/postulates/definitions/

E.g., Boolean Algebra

- AND, OR, NOT

- $\{0, 1\}$

- Closure, Commutative, Associative, Distributive, Identity, Inverse

Relational Model: Algebra: Solid Mathematical Model

Algebra:

- A set of operations

- A set of elements

- A set of axioms/postulates/definitions/

E.g., Relational Algebra

- Data Definition Language (DDL): To create relations

 - CREATE TABLE, INDEX, ..**

- Data Manipulation Language (DML): To modify relations

 - INSERT, UPDATE, DELETE, SELECT**

Relational Model: Formal vs. Informal Terminology

Relation

Schema	<u>Title</u>	ReleaseDate	RunningTime	Genre
Tuple 1	2001: A Space Odyssey	English	1 42	SciFi
Tuple 2	Rosemary's Baby	English	1 36	Horror
Tuple 3	The Birds	English	1 19	Horror
...	...	...	...	

Relation vs. Table

Tuple vs. Row

Domain vs. Column

Relational Model: Everything is Relation

Entity Set $\rightarrow$ Relation

Relationship Set $\rightarrow$ Relation

Data Modeling
in
RDBMS

Real World Entity

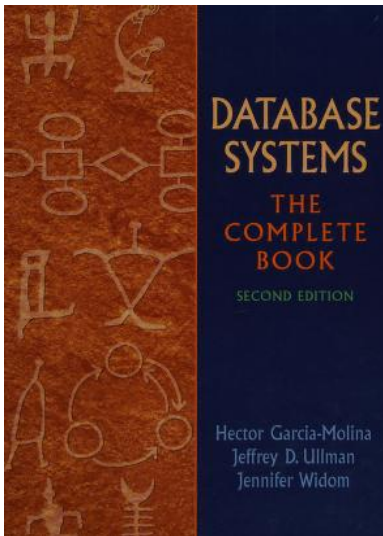
Conceptual Level | Entity Set & Relationship Set

Conceptual Level | Logical Level | Relation

Conceptual Level | Logical Level | Physical Level | SQL

Conceptual Level | Logical Level | Computable Entity

Outline: W04A



CH02 (2nd Ed.)

I) Relational Modeling

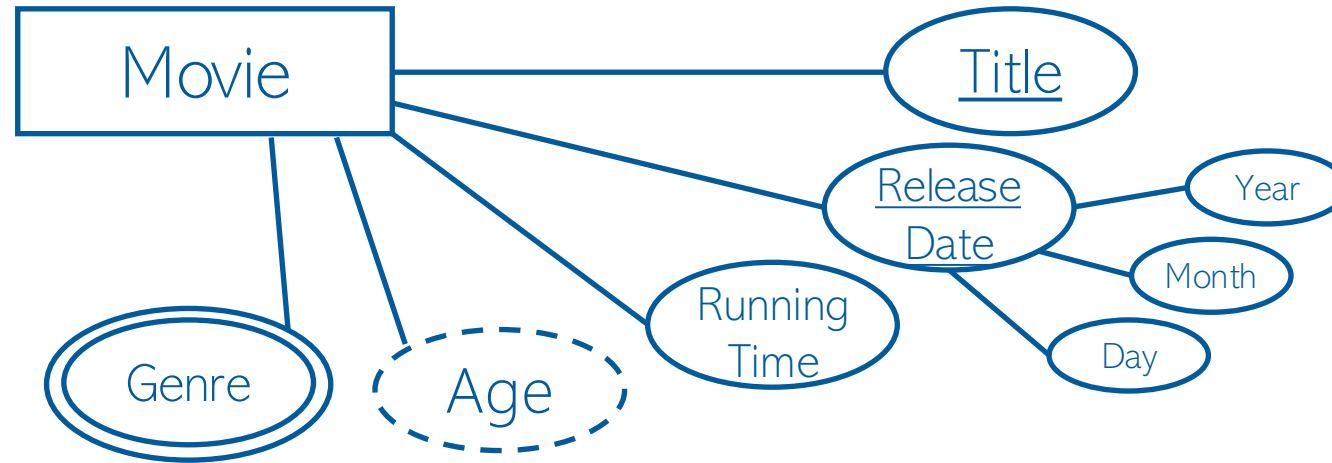
II) Entity to Relation

III) Relationship to Relation

IIII) IS-A to Relation

Hands-On

Relational Model: ER to Relation Transformation



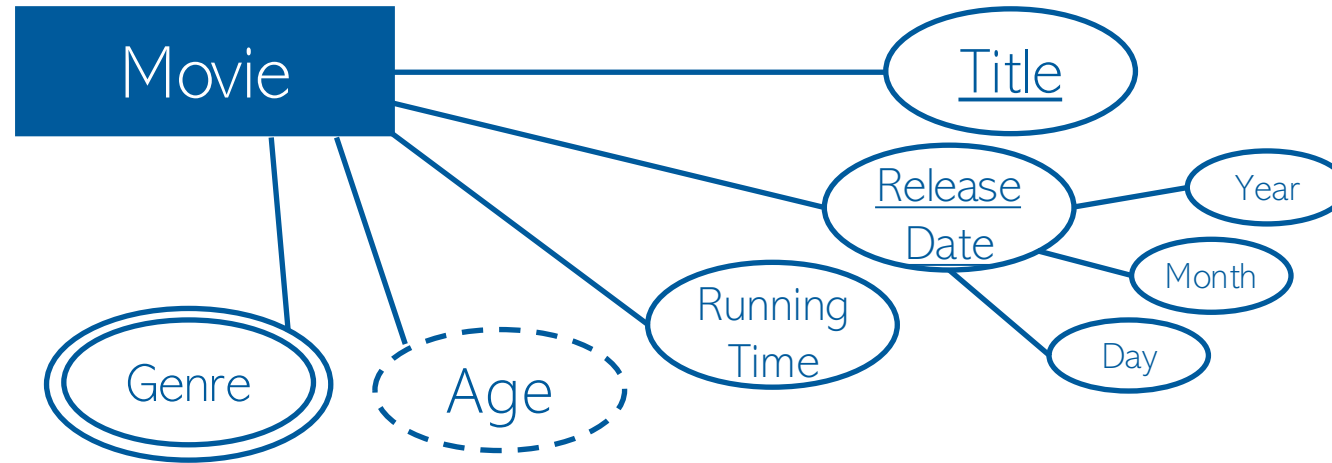
R1: Movie(Title, ReleaseYear, ReleaseMonth, ReleaseDay, RunningTime, Age)

R2: Genre(Title)

R3: MovieGenre(Movie.Title, Movie.ReleaseYear, Movie.ReleaseMonth, Movie.ReleaseDay, Genre.Title)

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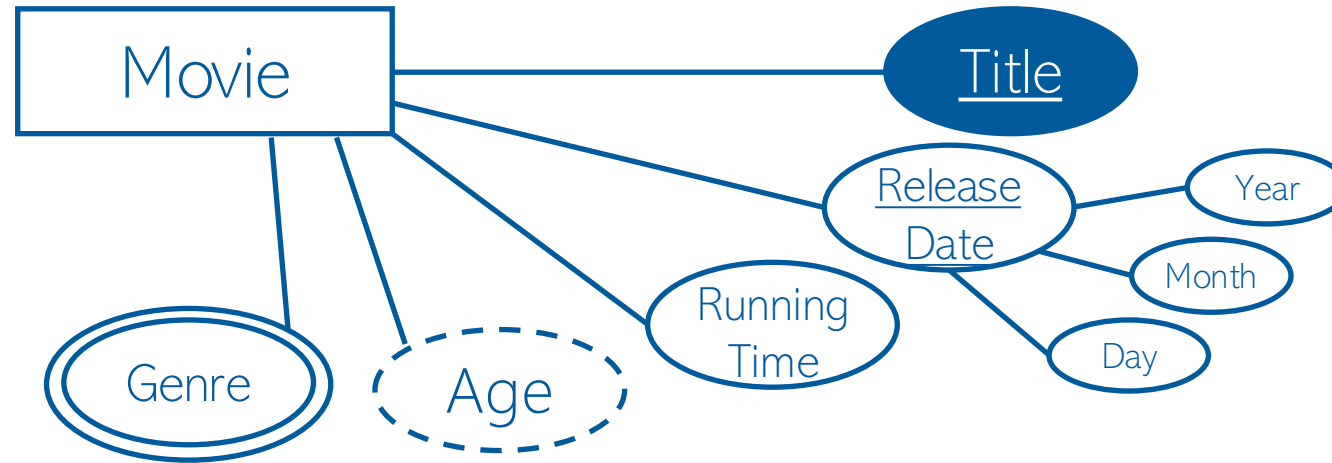
Relational Model: ER to Relation Transformation



R1: Movie()

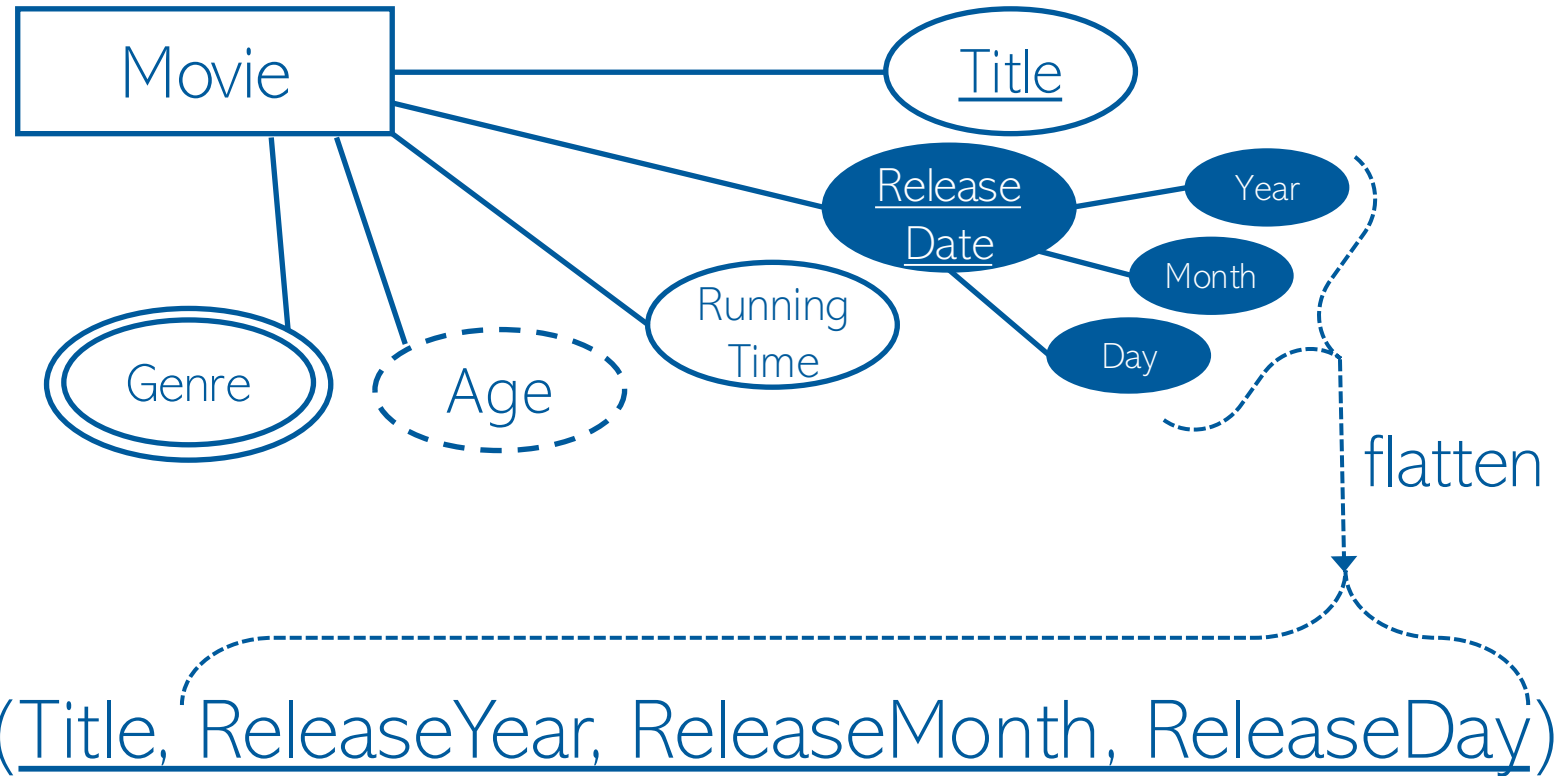
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Relational Model: ER to Relation Transformation

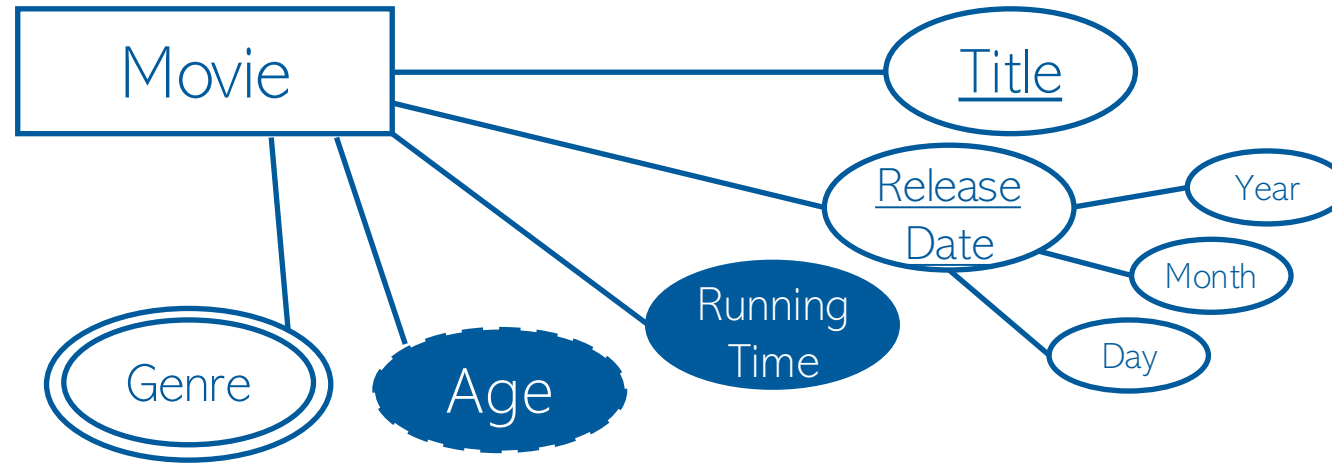


R1: Movie(Title)

Relational Model: ER to Relation Transformation

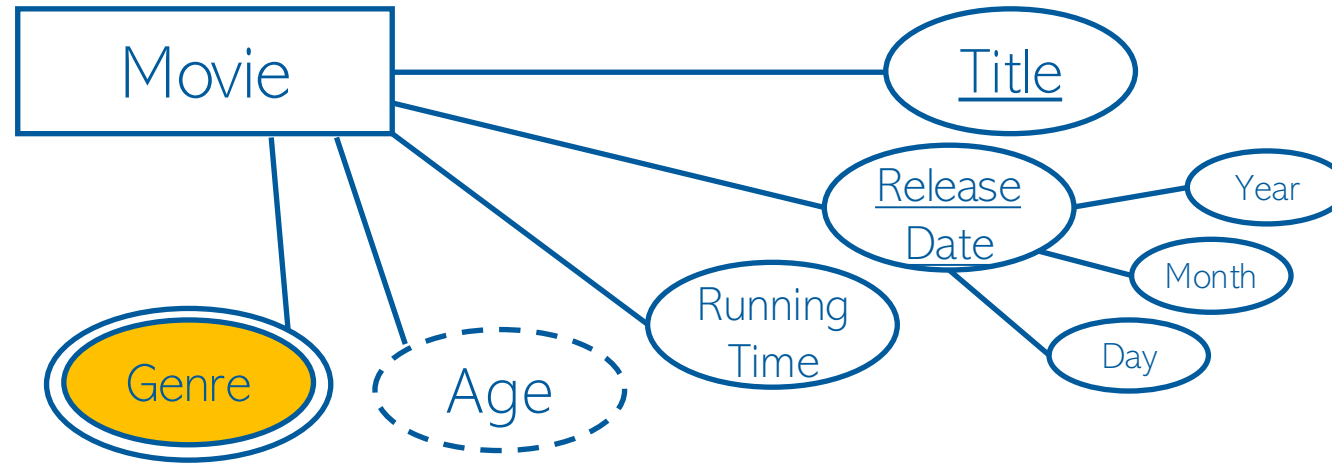


Relational Model: ER to Relation Transformation



R1: Movie(Title, ReleaseYear, ReleaseMonth, ReleaseDay, RunningTime, Age)

Relational Model: ER to Relation Transformation



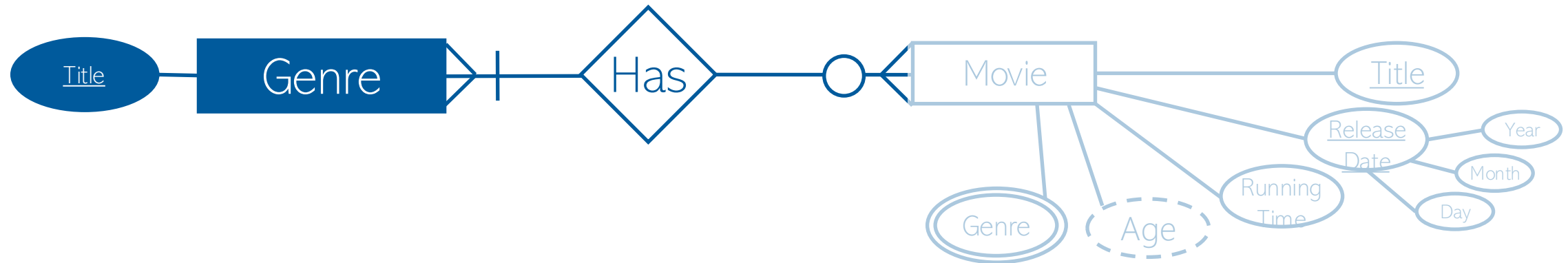
Relational model does not have multivalued attributes!

R1: Movie(Title, ReleaseYear, ReleaseMonth, ReleaseDay, RunningTime, Age)



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Relational Model: ER to Relation Transformation

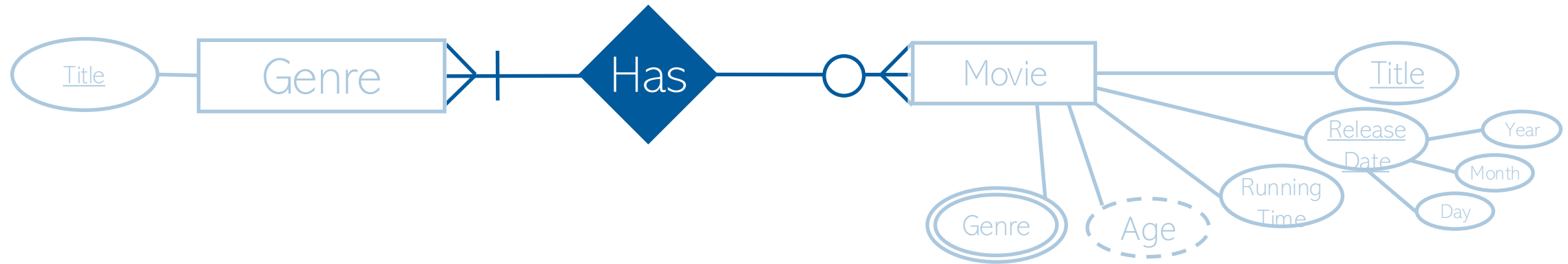


R1: Movie(Title, ReleaseYear, ReleaseMonth, ReleaseDay, RunningTime, Age)

R2: Genre(Title)

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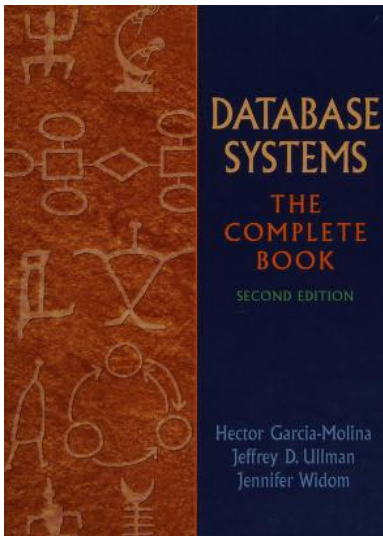
Relational Model: ER to Relation Transformation



R1: Movie(Title, ReleaseYear, ReleaseMonth, ReleaseDay, RunningTime, Age)

R2: Genre(Title)

Outline: W04B



CH02 (2nd Ed.)

I) Relational Modeling

II) Entity to Relation

III) Relationship to Relation

IIII) IS-A to Relation

Hands-On

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Outline: W04B

